COURSE 1

ZERO-TO-JOB-READY DATA ANALYST 2026

M00 | FOUNDATION

Beginner-First Learner Book

QUESTION -> MEASURE -> GRAIN -> TRUST -> EVIDENCE

Option B institute family | Course 1 learner edition

Not a learner release until module QA and Course 1 release gates pass

This book is your teacher

You are not expected to arrive with a technical background. This Foundation book teaches the analyst way of thinking before any heavy tool work begins. Read it in order. When a technical or business word first appears, the lesson defines it before asking you to use it.

THE RULE FOR EVERY LESSON

Understand the idea -> watch only when a useful visual is assigned -> follow the teacher once -> solve a changed small task alone -> validate -> explain it back -> save evidence. A protected review stays closed until a genuine attempt exists.

What this book does	What this book does not do
Explains from zero, gives a mental picture, demonstrates the full reasoning, shows expected results and beginner mistakes, and tells you exactly how to recover.	It does not make a video or a timer the mastery condition. It does not ask you to perform a skill that has not been taught. It does not unlock answers before an attempt.

Your normal route in this module:

LEARN	FOLLOW	DO ALONE	VALIDATE	TEACH BACK	SAVE EVIDENCE

After the nine lessons, complete the cumulative Foundation practice. Review opens only after a saved attempt. Gate A is the normal assessment. A clean Gate A pass ends the Gate. Gate B exists only after a failed A plus targeted repair and explicit assignment; Gate C only after a failed B plus another repair and explicit assignment.

ONE NEXT ACTION

Do not browse the module folder randomly. When the Mentor is available, follow its one-next-action instruction. In this book, the end of every lesson also names the exact next route.

Contents / Index

Every lesson is listed in the canonical order. Page numbers below are materialized from the rendered RC1 book; lesson titles are clickable in Word/PDF viewers that preserve internal links.

Module prerequisite & tool map

STARTING STATE

No technical prerequisite. You do not need Excel formulas, SQL syntax, Power BI or Python to begin M00. You need only basic file handling, a place to save evidence, and willingness to explain your reasoning in plain English.

Lesson block	What you build	Tool needed	Evidence
O1-O3 Orientation	Study route + safe workspace + artifact routing	File Explorer + text/document editor	route note, folder proof, routing table
F1-F2 Framing	decision-support workflow + bounded question	Writing only	workflow brief + question card
F3-F4 Measure & grain	metric card + source/result grain	Writing only	metric definition + grain map
F5 Trust	five data-quality checks	Writing only	quality checklist
F6 Integration	complete four-section Analyst Brief	Writing only	brief + validation explanation

Supporting files used after lesson teaching:

- `C1_M00_FOUNDATION_BEGINNER_FIRST_PRACTICE_BOOK_RC1.pdf` - guided-to-independent cumulative practice; attempt-first.
- `C1_M00_FOUNDATION_PRACTICE_REVIEW_PROTECTED_RC1.pdf` - opens only after a genuine saved cumulative-practice attempt.
- `C1_M00_FOUNDATION_GATE_A_RC1.pdf` - normal protected competency assessment.
- Gate B/C and their reviews - protected retest routes only after the required failure, repair and explicit assignment.

How to study, practise and take the Gate

READ WHY	LEARN WORDS	FOLLOW DEMO	DO FRESH TASK	VALIDATE	EXPLAIN

For each lesson, use the same beginner-first loop. Do not rush past the validation and teach-back stages; they are where you discover whether you understand the skill or only recognize the words.

1. Read sections 1-5. Build the mental model and glossary before touching a task.
2. Use section 6 only when a useful source is assigned. Watch the mapped segment for the stated reason, then return to the book.
3. Do section 7 with the teacher demonstration visible.
4. Close the demonstration as much as practical and complete section 12 - the changed small version alone.
5. Use section 13 to validate. A result that merely "looks right" is not evidence.
6. Do section 14 aloud or in writing. If you cannot explain it, mark the competency uncertain.
7. Save your genuine attempt. Only then may protected review/repair open if needed.
8. Complete the changed fresh retry after repair. Save the new evidence append-only.

GATE RULE

Gate A is normal. Passing Gate A ends the Gate. Gate B can be assigned only after failed A + saved A attempt + review/targeted repair. Gate C can be assigned only after failed B + saved B attempt + second repair. There is no percentage-average shortcut around a critical competency failure.

VIDEO RULE

A video is visual support, not a hidden prerequisite. If the external source is unavailable, the learner book must still teach the skill completely enough to continue.

O1 | How This Course Works

Foundation route | 35-50 min | Mastery is evidence, not time spent

1. Why this lesson matters

Before you learn Excel, SQL, Power BI or Python, you need to know how this course decides whether you are ready to move on. Otherwise it is easy to confuse reading a page with being able to do the skill.

2. What you are learning

- Distinguish Completed from Mastered.
- Use the attempt-first learning loop without opening answers too early.
- Understand evidence, repair and changed retry.
- Know why study time can guide planning but can never create a pass by itself.

3. What you already need / prerequisite

None. This is your orientation lesson. You only need this book and a place to save your course evidence.

4. Picture it in your head - simple real-world mental model

LEARN	FOLLOW	TRY ALONE	VALIDATE	EXPLAIN	EVIDENCE

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Completed	You finished the assigned reading, viewing or activity.
Mastered	You can do a changed version independently, validate it and explain why it is correct.
Evidence	A saved artifact that proves what you did, such as a brief, workbook, query, screenshot or explanation.
Review	Protected answer/explanation material that opens only after a genuine attempt.
Remediation	Targeted repair of the exact weak skill instead of restarting everything.
Fresh retry	A changed task that tests the same skill without letting memory of an answer become the pass.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Classify three learner situations.

1. Case A: You read the lesson and can repeat the definition. Status = Completed, not Mastered.
2. Case B: You attempt a changed task, validate it, and explain the result without the review open. Status = Mastered for that competency.
3. Case C: Your attempt fails because you confuse source grain and result grain. Route = save attempt -> open targeted review -> repair grain only -> close review -> solve a changed grain task.

8. What you should see / be able to say

- You can say: 'Completed means I finished the learning activity. Mastered means I proved the skill on a changed task.'
- You can draw the route: Learn -> guided attempt -> independent attempt -> validate -> explain -> evidence -> Gate.
- You know a clean Gate A pass ends the Gate; B and C are not extra practice for someone who already passed A.

9. Why it worked

The system separates exposure from proof. A learner can spend many hours watching lessons and still be unable to perform independently. Changed tasks and validation make mastery observable.

10. Common beginner mistakes

- Marking Mastered because the lesson felt easy.
- Opening the review before saving a genuine attempt.
- Taking Gate B or C after passing Gate A.
- Restarting an entire module because one small competency failed.
- Treating a timer or total hours as a pass condition.

11. If you get stuck - recovery ladder

1. Say the route aloud using only six words: learn, try, check, explain, save, test.
2. Look at the mental model above and point to the first step you have actually completed.
3. If you are unsure whether something is 'Mastered', ask: could I do a changed version with the lesson closed?
4. If still unsure, mark it Completed only. Mastery can be earned later; it should never be guessed.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Write the six-step study route from memory. Then classify these without looking back: (1) watched video only,

(2) completed guided example but not independent task, (3) solved changed task and validated it, (4) failed changed task after a genuine attempt.

13. Validate your result

- Your route includes an independent attempt before review.
- Only situation 3 is Mastered.
- Situation 4 routes to targeted repair and a changed retry, not automatic failure of the whole module.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

In 45-60 seconds, explain Completed vs Mastered to a friend who has never used an online course. Include why a changed retry is necessary.

15. Protected review + changed fresh retry

Save your answer first. Then open the protected O1 practice review only if assigned by the Mentor or if your validation is uncertain. After reading the repair, close it and solve a changed classification case; do not copy wording from the review.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save a short note named `O1_course_route.txt` or an equivalent entry in your evidence log. Master when you can reconstruct the route and classify a fresh situation correctly. Next: O2 - Files, Folders and Safe Data Handling.

O2 | Files, Folders and Safe Data Handling

Foundation route | 45-60 min | Mastery is evidence, not time spent

1. Why this lesson matters

Analysts often receive files that should never be changed directly. A safe folder habit protects the original data, makes your work reproducible and gives you a clear place for outputs and evidence.

2. What you are learning

- Create a simple analyst workspace.
- Distinguish source/raw data from working data.
- Use copy-not-move when preparing a working file.
- Name and save outputs so you can prove what changed without damaging the source.

3. What you already need / prerequisite

O1. You should know that evidence is saved work and that mastery requires a changed independent task.

4. Picture it in your head - simple real-world mental model

RAW SOURCE	**COPY**	**WORK AREA**	**OUTPUT**	**EVIDENCE**

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Raw data	The original data as received. Treat it as read-only unless the owner explicitly authorizes a change.
Working copy	A separate copy used for cleaning, exploration or transformation.
Output	A file or result produced by your analysis.
Path	The location of a file or folder on the computer.
Read-only habit	A working rule: do not edit or overwrite the original source file.
Reproducible	Someone can understand what you used and how your result was produced.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Build a safe Analyst_Lab workspace.

1. Create a folder named `Analyst_Lab`.
2. Inside it create: `raw_data`, `processed_data`, `outputs`, `screenshots`, and `documentation`.
3. Create or use a fictional file called `Orders.csv`. Put the original in `raw_data`.
4. Copy it into `processed_data` before making any practice change. Do not move or rename the raw original.
5. Save one screenshot of the folder structure in `screenshots`, and a one-line note in `documentation` saying what the raw file represents.

8. What you should see / be able to say

- `raw_data/Orders.csv` still exists unchanged.
- A separate working copy exists in `processed_data`.
- Your folders make the route visible even before you learn a technical tool.

9. Why it worked

Separating source, work and output prevents accidental overwrite and makes later troubleshooting easier. If your working file becomes wrong, you can return to an untouched source instead of guessing what the original looked like.

10. Common beginner mistakes

- Moving the only source file into a working folder.
- Editing directly inside `raw_data`.
- Saving screenshots, outputs and source files in one mixed folder.
- Using names such as `final2_reallyfinal.xlsx` with no meaningful structure.
- Deleting an unusual source row before investigating why it looks unusual.

11. If you get stuck - recovery ladder

1. Open File Explorer and locate the folder you are actually using; do not create folders in an unknown location.
2. Create only `Analyst_Lab` first, then one child folder at a time.
3. If you cannot tell whether you copied or moved a file, confirm that the original still exists in `raw_data` before continuing.
4. If a source was accidentally overwritten, stop. Restore from a known copy before doing more analysis.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Create a fresh workspace called `Support_Case_Lab` with the same safe structure. Put a fictional `Tickets.csv` in the raw folder, make a working copy, and write one sentence describing why the raw copy must stay unchanged.

13. Validate your result

- The raw file and working copy are two separate files.
- Your output/evidence locations are separate from source data.
- You can explain which file you would delete first if you wanted to restart your practice: the working copy, never the only raw source.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain the difference between raw data and a working copy. Give one example of an accidental action that the folder structure protects you from.

15. Protected review + changed fresh retry

Save the fresh `Support_Case_Lab` structure first. If assigned, open O2 review only afterward. Close the review and rebuild the structure using a different fictional dataset before claiming mastery.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save a screenshot or folder-tree text proving both raw and working copies exist. Master when you can build the structure without instructions and explain the copy-not-move rule. Next: O3 - Using the Learning System.

O3 | Using the Learning System

Foundation route | 35-50 min | Mastery is evidence, not time spent

1. Why this lesson matters

Course 1 contains learner books, practice, protected reviews, Gates and later projects. Knowing which artifact to open next prevents answer leakage and removes the 'which file do I use?' confusion that made earlier packages hard to study.

2. What you are learning

- Recognize the purpose of Lesson Book, Practice, Review, Gate and Repair artifacts.
- Choose the next artifact from your learning state.
- Keep protected answers closed until a saved attempt exists.
- Use the Mentor as a route guide rather than browsing folders randomly.

3. What you already need / prerequisite

O1 and O2. You should understand attempt-first mastery and safe evidence storage.

4. Picture it in your head - simple real-world mental model

LESSON BOOK	**PRACTICE**	**SAVE ATTEMPT**	**REVIEW IF NEEDED**	**FRESH RETRY**	**GATE**

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Learner book	The teacher-like book that explains and demonstrates the skill.
Practice artifact	The place where you follow, then solve a fresh task.
Protected review	Answers, reasoning or repair guidance hidden until a genuine attempt is saved.
Gate	A competency assessment used to decide whether you can move forward.
Route	The exact next artifact/action your Mentor tells you to use.
Unlock	Permission to open a protected item after its conditions are met.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Choose the correct next file.

1. Learner has not studied F3 yet -> open the learner book at F3, not the review.
2. Learner studied F3 and followed the teacher example -> open the F3 practice task and attempt it independently.
3. Learner saved a genuine F3 attempt but is uncertain -> review may open; repair only the weak concept.
4. Learner passed the fresh retry -> continue through the module and later take Gate A when assigned.
5. Learner failed Gate A -> Mentor assigns targeted repair; only after that can Gate B be assigned.

8. What you should see / be able to say

- You can choose one next artifact without browsing five folders.
- You understand that 'available on disk' does not mean 'allowed to open now'.
- You know that review and retest access is controlled by learning state, not curiosity.

9. Why it worked

A learning system is useful only when its files have roles. The route preserves assessment integrity: teaching is open, proof is attempt-first, and protected answers appear only when they help repair an already-attempted skill.

10. Common beginner mistakes

- Opening every PDF at the beginning of the module.
- Using a Gate review as study notes before attempting the Gate.
- Treating Gate B/C as ordinary practice.
- Skipping evidence and relying on memory that you 'did it'.
- Ignoring the Mentor's one-next-action route and navigating by filename guesswork.

11. If you get stuck - recovery ladder

1. Ask: am I learning, practising, reviewing a saved attempt, repairing, or testing?
2. Match that state to exactly one artifact role.
3. If no attempt has been saved, keep any review/answer file closed.
4. If the Mentor and a file disagree, stop and log the route mismatch instead of inventing your own path.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

For six new situations, write exactly one next artifact: (a) first exposure, (b) guided example finished, (c) independent attempt saved and wrong, (d) repair finished, (e) Gate A passed, (f) Gate A failed. Use only Learner Book, Practice, Review/Repair, Continue, or Gate B Assigned.

13. Validate your result

- First exposure routes to Learner Book.
- Independent work routes to Practice before Review.
- A Gate A pass routes to Continue, not Gate B.
- A Gate A failure routes to repair first; Gate B is not immediate.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain why a protected review can physically exist in a folder but still be 'locked' for you.

15. Protected review + changed fresh retry

After saving the six-situation routing attempt, use O3 review if needed. Then close it and route a changed set of situations from scratch.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save your routing table or evidence-log entry. Master when all fresh states route correctly without opening protected material. Next: F1 - What Does a Data Analyst Actually Do?

F1 | What Does a Data Analyst Actually Do?

Foundation route | 55-75 min | Mastery is evidence, not time spent

1. Why this lesson matters

A beginner can easily mistake the analyst job for 'using Excel' or 'making dashboards'. Tools matter, but the job is to turn a business question into trustworthy evidence that helps someone make a decision.

2. What you are learning

- Describe the analyst role as a decision-support workflow.
- Separate the business problem from the tool used.
- Identify where clarification, validation and communication belong.
- Recognize that analysts support decisions; they do not automatically own the business decision.

3. What you already need / prerequisite

O1-O3. No Excel, SQL, Power BI or Python knowledge is required.

4. Picture it in your head - simple real-world mental model

PROBLEM	**CLARIFY**	**DATA**	**PREPARE**	**ANALYZE**	**VALIDATE**	**COMMUNICA TE**	**DECISION**

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Stakeholder	A person or team who needs, uses or is affected by the analysis.
Business question	A decision-focused question the analysis is meant to answer.
Analysis	The reasoning and calculations used to turn data into evidence.
Validation	Checks that test whether your result is trustworthy and answers the intended question.
Insight	A supported finding that matters to the business question.
Decision support	Evidence and explanation that help the responsible person choose an action.

6. Watch / visual source - when useful

OPTIONAL VISUAL COMPANION

Alex The Analyst - What Does a Data Analyst Do: Meetings, Data, Excel, SQL
URL: https://www.youtube.com/watch?v=mCSbYbXWmH0
Segment: Full video; use for role/workflow visualization only
WHY: Watch to see what analyst work looks like around meetings, data, tools and communication. Do not treat the video as a substitute for the workflow taught below.
AFTER WATCHING: Return here immediately and do the teacher demonstration; do not continue browsing videos.

VIDEO LINK: Open the verified visual source

Internal fallback: this lesson contains the complete concept, worked demonstration, expected result, recovery ladder and fresh independent task. The external video is not required for mastery.

7. Follow with me once - fully worked teacher demonstration

Turn a vague request into the analyst workflow.

1. Request: 'Sales look bad. Make me a dashboard.'
2. Problem: something about sales concerns the requester, but the decision is not yet clear.
3. Clarify: who will use the answer, what decision they need to make, compared with what, for which period/region/product, and what 'sales' means.
4. Data: identify the minimum fields and the grain needed for that bounded question.
5. Prepare + Analyze: only after the question is clear, clean/transform and calculate the relevant measures.
6. Validate: check row counts, definitions, time scope and reconciliation before presenting.
7. Communicate: state what the evidence supports, what it does not support, and the limitations.
8. Decision: the business owner chooses the action; the analyst supplies defensible evidence.

8. What you should see / be able to say

- You can explain the job without naming a software tool.
- You can place validation before communication.
- You can tell the difference between 'I calculated a number' and 'I supported a decision with trustworthy evidence'.

9. Why it worked

The workflow starts with the decision and ends with evidence. This prevents tool-first work that looks polished but answers the wrong question. Validation before communication reduces the risk of confidently presenting a wrong result.

10. Common beginner mistakes

- Starting with 'Which tool should I use?' before clarifying the question.
- Assuming the stakeholder's suspected cause is true.
- Skipping validation because the chart looks reasonable.
- Claiming the analyst decides business policy automatically.
- Calling any interesting number an insight even when it does not affect the decision.

11. If you get stuck - recovery ladder

1. Ignore tools and write only: Who needs the answer? What decision? What comparison? What scope/time?

2. Draw the eight workflow boxes and put the request under the first box.
3. If you cannot decide what data is needed, the analytical question is probably still too vague.
4. If a result feels ready, ask what check would prove the number is not obviously wrong before communicating it.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh case: A support manager says, 'Customers are unhappy. Tell me what's going on.' Write one sentence for each of the eight workflow stages. Do not choose a tool and do not claim a cause that the case did not provide.

13. Validate your result

- Your first two stages identify the problem and missing decision context.
- Your data stage follows from the question rather than listing every available field.
- Validation appears before communication.
- Your final statement says the analysis supports a decision; it does not pretend the analyst owns it.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain the analyst job in 60-90 seconds without using the words Excel, SQL, Power BI or Python. Include the role of validation.

15. Protected review + changed fresh retry

Save your support-manager workflow first. If uncertain, open the F1 protected review after the attempt. Close it and repeat the workflow with a different business request before marking mastery.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save `F1_analyst_workflow_brief` in your evidence area. Master when you can map a changed vague request through all eight stages and defend why validation comes before communication. Next: F2 - From Vague Request to Analytical Question.

F2 | From Vague Request to Analytical Question

Foundation route | 60-80 min | Mastery is evidence, not time spent

1. Why this lesson matters

Real requests are often vague: 'sales are down', 'complaints are high', 'customers are leaving'. If you analyse immediately, you may answer a different question from the one the stakeholder actually needs.

2. What you are learning

- Separate a concern from a decision and an analytical question.
- Ask targeted clarification questions about decision, comparison, scope, time and meaning.
- Write a bounded analytical question that does not assume a cause.
- Separate confirmed facts from assumptions.

3. What you already need / prerequisite

F1. You should understand the analyst workflow and the idea of decision support.

4. Picture it in your head - simple real-world mental model

VAGUE CONCERN	CLARIFY 5 THINGS	BOUNDED QUESTION	DATA NEEDED

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Concern	The business issue or worry stated by the stakeholder.
Decision	The action or choice the stakeholder expects the analysis to inform.
Scope	The boundaries of the analysis: product, region, channel, team, customer group or other slice.
Comparison	What the result should be compared with: prior period, target, segment, benchmark or another baseline.
Bounded question	A question precise enough to guide data selection and analysis without pretending unknown facts are known.
Assumption	A provisional belief used only when necessary and kept visible until confirmed.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Clarify 'Sales are down' without guessing.

1. Concern: 'Sales are down.'
2. Decision: ask what decision the requester needs to make. Pricing? Staffing? Inventory? Campaign spend?
3. Comparison: down versus last month, last year, plan, or another region?
4. Scope: total company or a product/region/channel?
5. Time: which exact reporting period and date basis?
6. Meaning: does 'sales' mean orders, units, gross revenue, net revenue, recognized revenue, or something else?
7. Bounded question example after clarification: 'For completed online orders in North Region during August 2026, which product categories had the largest net-sales decline versus August 2025?'
8. Facts vs assumptions: write supplied facts separately from any temporary rule that still needs approval.

8. What you should see / be able to say

- Your question contains a measurable subject, scope, time basis and comparison.
- It does not contain an invented cause such as 'because prices rose'.
- You can point to the business decision the answer will support.

9. Why it worked

Clarification converts an emotional or vague concern into a testable question. A bounded question tells you what data is relevant and also tells you what conclusions are outside scope.

10. Common beginner mistakes

- Changing the request into a cause claim: 'Why did price increases reduce sales?' when no price effect is established.
- Asking many generic questions that do not affect the decision.
- Leaving words like 'recently', 'high' or 'sales' undefined.
- Choosing the dataset before knowing the question.
- Hiding an assumption inside the final wording as if it were confirmed.

11. If you get stuck - recovery ladder

1. Write five prompts: Decision? Comparison? Scope? Time? Meaning?
2. Answer only what the scenario actually supplies; put a question mark beside missing items.
3. Remove words that imply a cause unless the scenario provides causal evidence.
4. Read the question and ask: could two analysts choose different periods or definitions? If yes, it is still under-specified.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh case: 'Complaints are too high lately. Find the problem.' Write (1) the concern, (2) the decision you need to clarify, (3) five useful clarification questions, (4) one bounded analytical question using clearly labelled provisional assumptions only where unavoidable, and (5) facts vs assumptions.

13. Validate your result

- Your clarification questions cover decision, comparison, scope, time and meaning.
- Your bounded question can be answered with data in principle.
- It does not claim the cause of complaints.
- Unknown definitions remain questions/assumptions instead of invented policy.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain the difference between a business concern and an analytical question, then give one example of a dangerous hidden assumption.

15. Protected review + changed fresh retry

Save the complaint-case attempt first. If assigned, open F2 review, repair only the weak clarification element, then close it and rewrite a fresh bounded question for a new domain.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save your bounded-question card. Master when a changed vague request becomes a decision-linked question without hidden assumptions. Next: F3 - Metrics and KPIs.

F3 | Metrics and KPIs

Foundation route | 70-90 min | Mastery is evidence, not time spent

1. Why this lesson matters

A label such as 'conversion rate' or 'on-time delivery' is not a complete metric definition. If two analysts use different eligible records, dates or business rules, both calculations can look correct while disagreeing.

2. What you are learning

- Distinguish metric, KPI and target.
- Define rates using numerator and eligible denominator.
- Specify unit, time basis, includes/excludes and definition owner.
- Refuse to invent a business rule when the rule is unknown.

3. What you already need / prerequisite

F2. You should be able to state the decision and bounded analytical question before choosing a measure.

4. Picture it in your head - simple real-world mental model

DECISION	METRIC DEFINITION	ELIGIBLE BASE	CALCULATE	COMPARE

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Metric	A defined measurement used to describe performance or behaviour.
KPI	A metric that is especially important to a specific objective or decision.
Target	A desired or approved level for a metric; it is not the metric itself.
Numerator	For a rate, the count/value that meets the condition of interest.
Denominator	For a rate, the eligible base against which the numerator is compared.
Eligibility	The business rule deciding which records belong in the denominator.
Time basis	The date/timestamp rule used to assign records to a reporting period.
Definition owner	The role responsible for approving or confirming the

New word	Plain-English meaning
	business definition.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Build an On-Time Delivery Rate definition card.

1. Metric name: On-Time Delivery Rate.
2. Metric or KPI? It can be a KPI if timely delivery is central to the team's objective; KPI status comes from business importance, not a special formula.
3. Business meaning: share of eligible deliveries completed within the approved on-time rule.
4. Conceptual calculation: eligible deliveries meeting the approved on-time condition / all eligible deliveries.
5. Numerator: eligible deliveries classified on time.
6. Denominator: all deliveries eligible under the approved rule.
7. Unit: percentage.
8. Time basis: must be confirmed - for example actual delivery/completion date, order date, or another approved basis.
9. Includes/excludes: clarify cancelled, failed, rescheduled, customer-unavailable and test deliveries.
10. Target: include only if supplied or approved.
11. Definition owner: Delivery/Service Operations or another named business owner must confirm the rule.

8. What you should see / be able to say

- Two analysts following the same approved card should know what belongs in the calculation.
- You can calculate a simple rate: 80 on-time / 100 eligible = 0.80 = 80%.
- You can explain why 90% target and On-Time Delivery Rate are different concepts.

9. Why it worked

A metric card makes the hidden business choices visible. Numerator and denominator stop a rate from floating without an eligible base, while time basis and exclusions stop period/eligibility drift.

10. Common beginner mistakes

- Using the target as the metric definition.
- Writing 80% without knowing the eligible denominator.
- Inventing 'within 30 minutes' because an internet example used it.
- Calling every metric a KPI.
- Forgetting that counts/currency metrics may not need numerator/denominator in the same way a rate does.

11. If you get stuck - recovery ladder

1. Start with plain English: what event or amount am I measuring?

2. If it is a rate, ask 'out of exactly which eligible records?'
3. Write unknown policy as a question instead of filling it in.
4. Check whether the reporting period depends on order date, completion date or another timestamp.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh case: define a `Resolved Within SLA Rate` for a support team when the SLA threshold and reopened-ticket rule have not yet been supplied. Build a complete definition card without inventing those rules.

13. Validate your result

- Your numerator and denominator use the same eligibility universe.
- Unknown SLA/reopen rules are explicit questions, not guesses.
- Your time basis is stated or marked for confirmation.
- KPI status is linked to the business objective, not assumed from the metric name.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain metric vs KPI vs target to a beginner. Then explain numerator and denominator using a 20-out-of-25 example.

15. Protected review + changed fresh retry

Save the support-rate definition first. Use F3 review only afterward if assigned. Close it and build a fresh metric definition in another domain with at least one different eligibility rule.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save one complete metric definition card. Master when you can define a fresh rate so another analyst could reproduce it after the business rules are approved. Next: F4 - Grain and Required Data.

F4 | Grain and Required Data

Foundation route | 70-90 min | Mastery is evidence, not time spent

1. Why this lesson matters

Many analysis errors begin before any formula or query: the analyst misunderstands what one row represents. Repeated IDs can be correct, and combining tables at different grains can accidentally multiply records and inflate results.

2. What you are learning

- State source grain in plain English.
- Distinguish primary-row identity from legitimate repetition on a finer-grain table.
- State intended result grain before analysing.
- Identify the minimum data needed for a bounded question.

3. What you already need / prerequisite

F2-F3. You should have a bounded analytical question and a metric definition before deciding what rows and fields are required.

4. Picture it in your head - simple real-world mental model

QUESTION	SOURCE GRAIN	RESULT GRAIN	FIELDS NEEDED	CHECK

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Grain	What one row represents in a table or result.
Source grain	What one row represents in the data you receive.
Result grain	What one row or summary unit should represent in your answer.
Primary key	A field or combination intended to uniquely identify one row at that table grain.
Foreign key	A field that links to another table and can legitimately repeat on the many side.
Duplicate	A repeated record that violates the intended grain - not merely any repeated value.
Finer grain	A table with more detailed rows, such as one row per order-product instead of one row per order.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Why repeated OrderID can be correct.

1. Orders table: one row per Order. Grain = one order. OrderID should be unique there.
2. OrderLines table: one row per Order-Product line. Grain = one product line within an order. OrderID can repeat because one order can contain several products.
3. Suppose the question is monthly sales by region. Intended result grain = one Region-Month summary.
4. Before combining anything, write both source grains and the result grain.
5. Minimum data may include OrderID, region, date basis, line quantity/price/discount rules and status/eligibility - but only after the metric/question are defined.

8. What you should see / be able to say

- You no longer call every repeated OrderID a duplicate.
- You can write 'one row per ___' for each source and for the result.
- You can explain how an order-level amount could be multiplied if copied across several order-line rows without care.

9. Why it worked

Grain turns a table from a rectangle of values into a statement about real-world units. When source and result grains are explicit, you can detect whether joins, counts or rates would accidentally change the eligible unit.

10. Common beginner mistakes

- Deleting repeated customer or order IDs just because they repeat.
- Joining tables before stating their grains.
- Counting rows when the business unit should be distinct orders or customers.
- Using a finer-grain table without checking whether order-level values repeat.
- Listing every available field instead of the minimum fields required by the question/metric.

11. If you get stuck - recovery ladder

1. Point at one row and finish the sentence: 'This row represents one...'.
2. Ask which field should be unique at that grain.
3. If an ID repeats, ask whether the table is intentionally at a finer grain before calling it duplicate.
4. Write the intended result grain before deciding how to summarize.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh mini-tables: (A) one row per customer, (B) one row per support ticket, (C) one row per status event within each ticket. State each source grain, which IDs may repeat legitimately, and a result grain for 'weekly ticket resolution rate by team'. List the minimum fields you would need.

13. Validate your result

- CustomerID can repeat in tickets even if it is unique in customers.
- TicketID can repeat in ticket events because the event table is finer grain.
- Your result grain is a weekly team summary, not one row per event.
- Required fields follow the metric and date/eligibility rules.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain grain without using the word 'row' in your first sentence, then give one example where a repeated ID is expected rather than duplicate.

15. Protected review + changed fresh retry

Save the mini-table reasoning before opening F4 review. If repair is needed, close the review and solve a changed three-table grain problem before mastery.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save source-grain/result-grain statements and one duplication-risk note. Master when you can defend legitimate repetition and choose a result grain for a fresh question. Next: F5 - Data Quality.

F5 | Data Quality

Foundation route | 75-95 min | Mastery is evidence, not time spent

1. Why this lesson matters

Bad-looking data is not an instruction to delete or fill it. An analyst first checks whether the data is complete, unique at the correct grain, valid, consistent and timely enough for the decision.

2. What you are learning

- Use five beginner-friendly data quality dimensions.
- Write dataset-specific checks rather than generic slogans.
- Separate an anomaly signal from a confirmed error.
- Describe the business impact and next action for a failed check.

3. What you already need / prerequisite

F4. Quality checks depend on knowing what one row should represent and which fields/rules matter to the question.

4. Picture it in your head - simple real-world mental model

EXPECTED GRAIN	5 QUALITY CHECKS	INVESTIGATE	IMPACT	ACTION

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Completeness	Whether required values/records are present for the intended use.
Uniqueness	Whether the intended row identifier is unique at the stated grain.
Validity	Whether values obey expected formats, ranges or logical rules.
Consistency	Whether labels/definitions/units/timestamps follow compatible rules.
Timeliness	Whether the data is current and complete enough for the requested reporting period.
Anomaly	Something unusual that deserves investigation; unusual does not automatically mean wrong.
Imputation	Filling missing values using a rule or estimate. This must never be done silently.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Check a delivery dataset without auto-fixing.

1. Grain: intended one row per delivery; DeliveryID should be unique.
2. Completeness: count missing PromisedAt or DeliveredAt among records that should be completed/eligible.
3. Uniqueness: test duplicate DeliveryID at delivery grain; do not test whether Region repeats because regions should repeat.
4. Validity: DeliveredAt should not be before a logically required earlier timestamp unless the business process allows it; status must be in the approved status set.
5. Consistency: check labels such as 'North', 'NORTH', 'North Region' and timestamp/timezone rules.
6. Timeliness: confirm LastUpdatedAt covers the requested reporting cut-off.
7. For every failure: record what you observed, possible business impact and what must be confirmed before changing data.

8. What you should see / be able to say

- Your checks mention actual fields/rules rather than only saying 'check nulls'.
- You do not delete an unusual row until its grain/business meaning is understood.
- You can explain how a quality issue could bias a metric or make a period incomplete.

9. Why it worked

Quality is fitness for the intended analysis, not visual cleanliness. Dataset-specific checks connect a defect to the business question, making repair auditable instead of arbitrary.

10. Common beginner mistakes

- Treating every repeated value as a duplicate.
- Filling NULL with zero without meaning/approval.
- Deleting outliers because they look extreme.
- Running generic checks that ignore the metric's required fields.
- Ignoring data freshness when analysing a recent period.

11. If you get stuck - recovery ladder

1. Restate the table grain and required metric fields.
2. For each dimension ask one concrete question using a field name.
3. If a value looks wrong, write 'investigate' before writing 'fix'.
4. Ask what decision would be harmed if the issue were real.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh case: an appointment table is intended one row per appointment with AppointmentID, Clinic, ScheduledAt, CheckInAt, SeenAt, Status and LastUpdatedAt. Write one scenario-specific check for each of the five dimensions, plus likely impact and next action for any failure.

13. Validate your result

- Uniqueness is tested on AppointmentID, not on Clinic.
- Completeness focuses on fields required for the wait-time metric and eligible statuses.
- Validity/consistency checks are logically tied to timestamps/status labels.
- Timeliness refers to the requested cut-off, not simply 'data should be recent'.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Explain why 'missing', 'duplicate-looking' and 'unusual' are investigation signals rather than automatic delete/fill instructions.

15. Protected review + changed fresh retry

Save your five appointment checks first. Use F5 review only after the attempt. Then close it and write five changed checks for a different dataset before mastery.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save a five-dimension quality checklist with impact/action notes. Master when you can create fresh dataset-specific checks without auto-correcting. Next: F6 - Analysis Validation and the Complete Analyst Brief.

F6 | Analysis Validation and the Complete Analyst Brief

Foundation route | 90-120 min | Mastery is evidence, not time spent

1. Why this lesson matters

An analyst brief ties the whole Foundation module together. It shows the decision, metric, grain, required data, quality checks, validation evidence and Definition of Done before tool work begins.

2. What you are learning

- Build a coherent four-section Analyst Brief.
- Distinguish validation from reconciliation.
- Use assumptions and limitations without disguising them as facts.
- Write a testable Definition of Done and a Ready/Not Ready judgment.

3. What you already need / prerequisite

F1-F5. This lesson deliberately combines every Foundation competency. If one concept is weak, repair that lesson instead of skipping it.

4. Picture it in your head - simple real-world mental model

1 PROBLEM	2 MEASURE + DATA	3 TRUST CHECKS	4 DONE + DEFENSE

Read the boxes from left to right. The purpose is not to memorize a slogan; it is to keep the order of reasoning visible while you work.

5. New words before we use them - beginner glossary

New word	Plain-English meaning
Analyst Brief	A compact planning document that states the question, metric/data needs, trust checks and acceptance criteria.
Validation evidence	Checks showing the result behaves as expected and answers the intended question.
Reconciliation	Comparing a result or control total with a trusted reference under matching definitions, scope and time.
Sanity check	A quick reasonableness test that catches obviously impossible or implausible results.
Definition of Done	Testable acceptance criteria that state what must be true before delivery.
Limitation	A known boundary in the available data/method that affects what the analysis can support.
Ready / Not Ready	A reasoned judgment about whether analysis can begin

New word	Plain-English meaning
	responsibly with the current definitions/data.

6. Watch / visual source - when useful

BOOK-LED LESSON

No external video is required for this lesson. The mental model and worked demonstration below are the primary teaching route. A weak or random video has not been added simply to increase video count.

7. Follow with me once - fully worked teacher demonstration

Build the four-section brief for subscription cancellations.

1. PROBLEM: stakeholder = retention/operations; decision = where to investigate/intervene first; ask comparison/scope/time and what counts as cancellation; write a bounded question without claiming why customers cancel.
2. MEASURE + DATA: define cancellation rate only after eligibility and time basis are approved; state source grain (for example one row per subscription cycle) and intended result grain (for example region-plan-month). List only required fields/rules.
3. TRUST CHECKS: write completeness, uniqueness, validity, consistency and timeliness checks; add analysis validation such as eligible counts and rate bounds; reconcile only to a comparable trusted control under matching scope/definition.
4. DONE + DEFENSE: require approved definitions, traceable eligible counts, passed/understood quality checks, validated/reconciled outputs where comparable, documented assumptions/limitations, and a conclusion that does not exceed the evidence.
5. READY/NOT READY: if the cancellation definition or denominator is unknown, the analysis may be Not Ready for a final rate even if exploratory data inspection can begin.

8. What you should see / be able to say

- All four sections refer to the same business question and metric.
- Reconciliation never means forcing your result to equal a reference with a different definition.
- Your Definition of Done can be checked yes/no instead of saying 'analysis looks good'.

9. Why it worked

The brief creates coherence. It prevents a later tool from hiding an earlier definition error and gives you a checklist for deciding when evidence is trustworthy enough to communicate.

10. Common beginner mistakes

- Using a generic quality checklist unrelated to the chosen metric.
- Reconciling against an incomparable reference and changing data until totals match.
- Writing 'dashboard complete' as the Definition of Done.
- Hiding missing business rules inside assumptions.
- Calling the analysis Ready because the files opened successfully.

11. If you get stuck - recovery ladder

1. Write the four headings only: Problem; Measure & Data; Trust Checks; Done & Defense.

2. Under Problem, make the question bounded before filling the other sections.
3. Under Measure & Data, state metric rule and source/result grain before listing fields.
4. Under Trust Checks, ask what evidence would make you trust the numerator, denominator and reporting period.
5. Under Done, turn each important requirement into a yes/no acceptance criterion.

STOP RULE

Use the smallest hint that gets you moving. Do not jump straight to a finished answer. If you open protected review after a saved attempt, close it before the changed retry.

12. Now you do a small version alone

FRESH INDEPENDENT TASK

Fresh case: 'Subscription cancellations seem worse. Tell us which plans and regions need attention first.' Create a complete four-section Analyst Brief. No approved cancellation denominator, target or causal reason is supplied. You must expose those gaps rather than invent them.

13. Validate your result

- The bounded question does not claim a cause.
- Metric eligibility/time rules are approved or explicitly unresolved.
- Source grain and result grain are both stated.
- Five quality dimensions are scenario-specific.
- Reconciliation uses only a comparable trusted control and keeps definition differences visible.
- Definition of Done is testable and Ready/Not Ready is defended.

If any check fails, do not mark Mastered. Record which check failed and use the recovery ladder or protected review only after your genuine attempt is saved.

14. Teach it back in your own words

EXPLAIN-BACK

Give a 90-second explanation of the four-section Analyst Brief. Then explain the difference between validation and reconciliation with one example.

15. Protected review + changed fresh retry

Save the full brief before opening F6 review. If a competency fails, open only the relevant repair guidance, close it, and rebuild that part on a changed domain. Do not memorize the model brief.

16. Evidence / mastery + exact next route

FINISHED WHEN

Save the four-section Analyst Brief plus one short validation/reconciliation explanation. Master when a fresh end-to-end brief is coherent across question, metric, grain, quality and acceptance criteria. Next route: complete Foundation cumulative practice; after a genuine saved attempt and review/repair if needed, take Foundation Gate A when the Mentor assigns it.

End-of-module mastery checklist

Do not check a box because you remember seeing the topic. Check it only if you can perform a changed version, validate it and explain it without the protected answer open.

Competency	Mastery proof	Status
O1	Explain Completed vs Mastered; route failure to targeted repair + changed retry.	[ ] Evidence saved
O2	Create a safe raw/work/output/evidence folder structure without overwriting the source.	[ ] Evidence saved
O3	Choose the correct learner artifact from a fresh learning state and preserve review/Gate locks.	[ ] Evidence saved
F1	Describe the analyst job as a decision-support workflow with validation before communication.	[ ] Evidence saved
F2	Turn a vague concern into a bounded, decision-linked analytical question without hidden cause assumptions.	[ ] Evidence saved
F3	Define a metric/KPI clearly, including numerator/eligible denominator for rates and unresolved business rules.	[ ] Evidence saved
F4	State source/result grain, defend legitimate repeated IDs and identify minimum required data.	[ ] Evidence saved
F5	Write dataset-specific completeness, uniqueness, validity, consistency and timeliness checks without auto-fixing.	[ ] Evidence saved
F6	Build a coherent four-section Analyst Brief with validation/reconciliation, limitations, Definition of Done and Ready/Not Ready.	[ ] Evidence saved

CUMULATIVE PRACTICE ROUTE

After all nine lesson competencies are at least Completed and you have attempted the small independent tasks, open `C1_M00_FOUNDATION_BEGINNER_FIRST_PRACTICE_BOOK_RC1.pdf`. Save a genuine cumulative attempt before opening its protected review. Gate A comes only after the practice route is complete and the Mentor assigns it.

Revision / quick reference

FOUNDATION IN ONE SENTENCE

Clarify the decision -> define the question and metric -> state grain/data needs -> test data/result trust -> communicate only what the evidence supports.

Memory cue	What to remember
Analyst workflow	Problem -> Clarify -> Data -> Prepare -> Analyze -> Validate -> Communicate -> Support Decision
Clarify 5	Decision | Comparison | Scope | Time | Meaning
Metric rate	Numerator / eligible denominator; plus unit, time basis, include/exclude, owner
Grain	Write: one row represents one ____; also write intended result grain
Data quality 5	Completeness | Uniqueness | Validity | Consistency | Timeliness
Trust	Validate internal logic; reconcile only to comparable trusted controls; sanity-check plausible ranges
Analyst Brief	1 Problem | 2 Measure & Data | 3 Trust Checks | 4 Definition of Done + Defense
Assessment	Practice attempt -> review if needed -> fresh retry -> Gate A -> repair/B only after failure -> repair/C only after failure

Spaced fresh revision:

- D+1: rebuild one bounded question + one metric definition from memory.
- D+4: solve a mixed mini-case that asks for question, metric, grain and two quality checks.
- D+10: build a short Analyst Brief in a new domain without looking at the model.
- D+30: explain the full Foundation reasoning in interview style and defend one assumption/limitation.

FOUNDATION EXIT

You are ready to leave M00 only when cumulative practice and Gate A (or the explicitly assigned repair/retest route) show competency. Hours studied and pages read do not override missing evidence.